

Selcoperm test kit 98842487

Operating instructions



Other languages

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GRUNDFOS 

English (GB) Installation and operating instructions

Original installation and operating instructions

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Read this document before installing the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1. General instructions

1.1 Safety instructions

- Keep away from heat.
- Store in a cool place.
- Keep away from food, drink and animal feeding stuffs.
- Keep out of the reach of children. Reagents are designed for use in chemical analysis only.
- Wash hands after working with the test kit. Take off contaminated clothing immediately.
- Avoid contact with skin and eyes. Wear safety goggles and gloves.
- Do not eat, drink or smoke at the place of work.

CAUTION

Hazardous chemicals

- Minor or moderate personal injury
- Read the method description completely before performing the tests. Read the Material Safety Data Sheets (MSDS).
 - MSDS are available on:
net.grundfos.com/qr/i/98734551



2. Test kit content

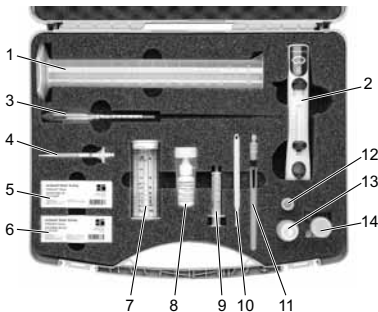


Fig. 1 Test kit content

Pos.	Description
1	Measuring cylinder, 250 ml
2	HandyStep S titrator
3	Hydrometer
4	Dispenser tip
5	Acidifying GP tablets, 100 pcs
6	Chlorine HR tablets, 100 pcs
7	Sampling container
8	Total hardness test kit
9	Syringe, 6 ml
10	Stirring rod
11	Thermometer
12	Neutral tube, 5 ml
13	Small bottle with 30 ml titrant (2N Sodium Thiosulphate solution ($\text{Na}_2\text{S}_2\text{O}_3$))
14	Test jar, crystal clear, 60 ml

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3. Checking the concentration of the NaClO solution

The HandyStep S titrator enables accurate titrimetric analysis in the field, using small quantities of sample and reagent consumables.

3.1 Required material

- HandyStep S titrator
- Dispenser tip
- Syringe, 6 ml
- Test jar, crystal clear, 60 ml
- Neutral tube, 5 ml
- Stirring rod
- Titrant: 30 ml of 2N sodium thiosulphate solution ($\text{Na}_2\text{S}_2\text{O}_3$)
- Chlorine HR tablets
- Acidifying GP tablets
- Softened water (from soft-water sample valve of the Selcoperm system)

HandyStep S titrator

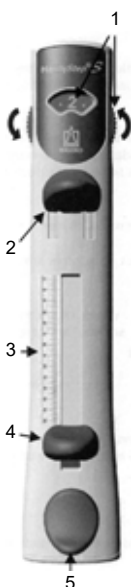
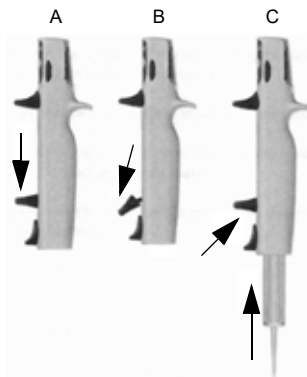


Fig. 2 HandyStep S titrator

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3.2 Fitting the dispenser tip to the HandyStep S titrator



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Fig. 3 Fitting the dispenser tip to the titrator

1. Hold the HandyStep S titrator (fig. 1, Pos. 2) vertically (titrating position). Push the locking/filling lever (fig. 2, Pos. 4) to its lower stop (A).
2. Push the locking/filling lever down gently until it inclines (B).
 - The locking bar inside the titrator opens.
3. Insert the top of the dispenser tip (fig. 1, Pos. 4) into the titrator. Pull the locking/filling lever back into its starting position (C).
 - The locking bar inside the titrator closes.

The dispenser tip is now held firmly inside the titrator ready to use.

Pos.	Description
1	Stroke setting wheel (slow/rapid dispense)
2	Dispensing lever
3	Titration scale (units dispensed)
4	Locking/filling lever
5	Syringe ejector

3.3 Filling and zeroing the HandyStep S titrator

1. Open the bottle with 2N sodium thiosulphate solution (titrant) (Fig. 1, Pos. 13).
2. Fill a small amount (approx. 2 ml) into the neutral tube (Fig. 1, Pos. 12) to prevent contamination of the solution.
3. Dip the end of the dispenser tip (Fig. 1, Pos. 4) into the titrant in the neutral tube.
4. Move the locking/filling lever (Fig. 2, Pos. 4) upwards until its upper stop.
 - Titrant is drawn from the neutral tube into the dispenser tip.
5. Make sure that no air is trapped inside the dispenser tip.
 - To remove air, hold the titrator at an angle and push and pull the locking/filling lever up and down a few times.
 - Move the locking/filling lever upwards until its upper stop (see Fig. 4).

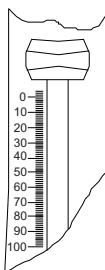


Fig. 4 Top position of locking/filling lever

- The dispenser tip is full of titrant.
6. Hold the end of the dispenser tip over a waste receptacle or tissue paper. Gently push the dispensing lever.
 - Titrant is discharged from the dispenser tip.
 7. Push the dispensing lever, until the bottom of the locking/filling lever is levelled with the zero mark on the titration scale (see Fig. 5).

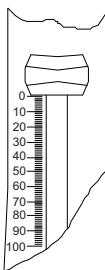


Fig. 5 Bottom of locking/filling lever levelled with zero mark

- A small quantity of titrant is discharged. The titrator is primed for titration.
8. Remove any excess titrant from the bottom of the dispenser tip carefully with a piece of clean tissue paper.
 9. Make sure that the stroke setting wheel (Fig. 2, Pos. 1) is set to position "1" (see Fig. 6).

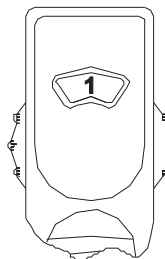


Fig. 6 Stroke setting wheel

- Setting the stroke setting wheel to a higher number increases the quantity of titrant dispensed per full depression of the dispensing lever.

The titrator is zeroed and ready to titrate.

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3.4 Titration steps

1. Take a NaClO sample from the Selcoperm system. See Selcoperm installation and operating instructions.



Measure the concentration immediately after extracting the sample.

2. Fill 20 ml of softened water into the test jar (Fig. 1, Pos. 14).



Do not touch the reagent tablets with the fingers.

Add the reagents in the correct sequence.

3. Add 2 Chlorine HR tablets (Fig. 1, Pos. 6) and pound them with the stirring rod.
4. Add 2 Acidifying GP tablets (Fig. 1, Pos. 5) and pound them with the stirring rod.
5. Mix well by swivelling carefully.
 - It is possible that small parts of the tablets are not dissolved. This has no effect to the test result.
6. Fill the 6 ml syringe (Fig. 1, Pos. 9) with 5 ml of NaClO solution sample.
 - Make sure that no air bubbles are drawn into the syringe.
7. Add the 5 ml of sample to the reagents in the test jar. Swivel carefully.
 - The colour of the sample turns into a reddish brown immediately.
8. Hold the HandyStep S titrator (Fig. 1, Pos. 2) in one hand and the test jar in the other hand. Insert the dispenser tip (Fig. 1, Pos. 4) into the test jar so that its end is submerged in the sample.
9. Push the dispensing lever (Fig. 2, Pos. 2) to add titrant, while swivelling the sample in the test jar.
 - The titrant is mixed homogeneously for highly accurate results.
10. Add titrant until the colour of the sample has turned from reddish brown to clear.



Stop adding titrant when the sample turns clear.

This marks the endpoint of the titration.

11. The position of the bottom of the locking/filling lever (Fig. 2, Pos. 4) at the titration scale (Fig. 2, Pos. 3) shows the number of units that have been dispensed.
 - Note down the number of units dispensed.

3.5 Calculation

1. Read the number of units dispensed at the titration scale (Fig. 2, Pos. 3).
2. Multiply the number of units dispensed by the conversion factor 0.16:
 - Units dispensed x 0.16 = value g/l chlorine

Units dispensed	Chlorine [g/l]
35	5.60
36	5.76
37	5.92
38	6.08
39	6.24
40	6.40
41	6.56
42	6.72
43	6.88
44	7.04
45	7.20
46	7.36
47	7.52
48	7.68
49	7.84
50	8.00
51	8.16
52	8.32
53	8.48
54	8.64
55	8.80

3.6 Removing the dispenser tip

1. Hold the end of the dispenser tip (Fig. 1, Pos. 4) over the neutral tube (Fig. 1, Pos. 12).
2. Push the locking/filling lever (Fig. 2, Pos. 4) to its lower stop.
 - Unused titrant is discarded.
 - Always use new titrant in the neutral tube in order to obtain a safe result at the next measurement.
3. Push the locking/filling lever down and forward to release the locking bar.
4. Push down the syringe ejector (Fig. 2, Pos. 5) and remove the dispenser tip.
5. Push the locking/filling lever back in its original starting position.

4. Checking the water hardness

A properly functioning water softener is crucial for the efficiency and longevity of the Selcoperm system. High levels of calcium carbonate in the electrolyte lead to a visible precipitation on the electrodes of electrolyser cell.

The water hardness test is a drop test. The hardness is determined by counting the drops of reagent, until the colour changes.

4.1 Water hardness test procedure

1. Take a sample from the soft water sample valve at the Selcoperm system. See Selcoperm installation and operating instructions.
2. Fill the sampling container (Fig. 1, Pos. 7) to the 5 ml mark.
3. Hold the reagent bottle vertically. Add the reagent drop by drop while swivelling the sample carefully, until its colour turns from red to green.
 - Count the number of drops. One drop equates to one degree of total water hardness.

4.2 Result

- If the sample turns green with one drop of reagent, the water is soft.
 - The water softener is functioning properly.
- If the sample turns red and requires more than one drop to turn green, the water is not soft.
 - The water softener needs regeneration or the resin beads must be replaced.

4.3 Conversion of water hardness

°d	°e	°f	mg/l CaO	mg/l CaCO ₃	mmol/l
1	1.3	1.8	10	18	0.18
2	2.5	3.6	20	36	0.36
3	3.8	5.4	30	54	0.54
4	5.0	7.1	40	71	0.71
5	6.3	8.9	50	89	0.89
6	7.5	10.7	60	107	1.07
7	8.8	12.5	70	125	1.25
8	10.0	14.3	80	143	1.43
9	11.3	16.1	90	161	1.61
10	12.5	17.8	100	178	1.78

5. Checking the specific gravity (SG) of the electrolyte

SG is the specific gravity of the electrolyte entering the electrolyser cell. See Selcoperm installation and operating instructions.

5.1 SG test procedure

1. Flush a clean and dry measuring cylinder (Fig. 1, Pos. 1) with electrolyte from the electrolyte sample valve.
2. Fill the measuring cylinder with a sample from the electrolyte sample valve.
3. Place the hydrometer (Fig. 1, Pos. 3) in the measuring cylinder.
4. Take the reading from the hydrometer.

6. Checking the temperature of the NaClO solution

1. Take a sample from the NaClO solution sample valve at the Selcoperm system.
2. Place the thermometer (Fig. 1, Pos. 11) in the sample and take the reading.
 - The temperature should not exceed 40 °C.

7. Spare parts

Description	Product number
Hydrometer	98504212
Thermometer	98708063
Stirring rod	95727771
Dispenser tip, 5 pcs	99619798
Total hardness test kit	98948745
Set of 100 Acidifying GP tablets and 100 Chlorine HR tablets	98144705
Set of 30 ml titrant (2N sodium thiosulphate solution (Na ₂ S ₂ O ₃)) and dispenser tip	99619819

8. Disposal

This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.

The used analysis sample can be flushed down the drain with tap water. Unused, expired reagent must be disposed of according to local regulations.

See also end-of-life information at www.grundfos.com/product-recycling.

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